

HAT_RASPI5_INA226 Fabrication Document

Layer Stack Legend

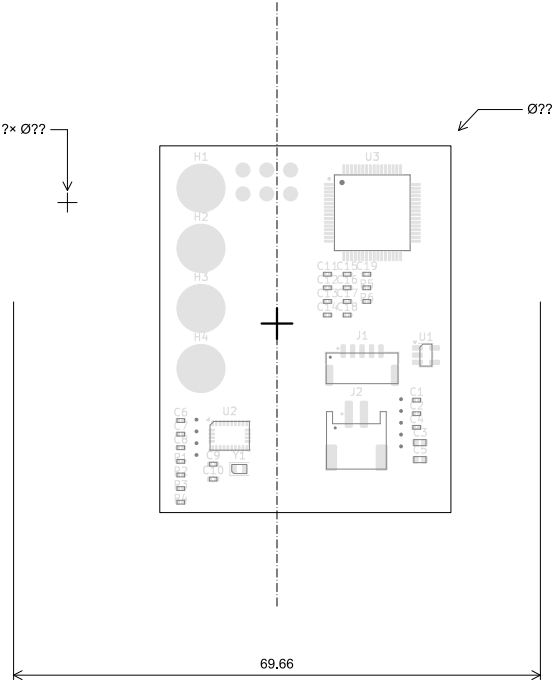
	Material	Layer	Thickness	Dielectric	Type	Gerber
		F,Paste			Paste Mask	
		F,Silkscreen			Legend	GBR
		F,Mask	0,02mm	Solder Resist	Solder Mask	GBR
	Copper	L1 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
	Prepreg		0,18mm	FR4_7628	Dielectric	
	Copper	L2 (GND)	0,035mm (1,00oz)		Plane	GBR
	Core		0,4mm	FR4	Dielectric	
	Copper	L3 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
	Prepreg		0,18mm	FR4_7628	Dielectric	
	Copper	L4 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
	Core		0,4mm	FR4	Dielectric	
	Copper	L5 (GND)	0,035mm (1,00oz)		Plane	GBR
	Prepreg		0,18mm	FR4_7628	Dielectric	
	Copper	L6 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
		B,Mask	0,02mm	Solder Resist	Solder Mask	GBR
		B,Silkscreen			Legend	GBR
		B,Paste			Paste Mask	

Total thickness: 1,66mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0,2032	0,28	L2

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.

B. Tg 170 C OR EQUIVALENT.

C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY DATALOGIC VIETNAM.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE38,500 × 48,500 mm

BOARD THICKNESS1,660 mm

TRACE WIDTHN/A mm

TRACE TO TRACE0,200 mm

MIN. HOLE (PTH)3,200 mm

MIN. HOLE (NPTH)N/A mm

ANNULAR RING1,600 mm

COPPER TO HOLE0,254 mm

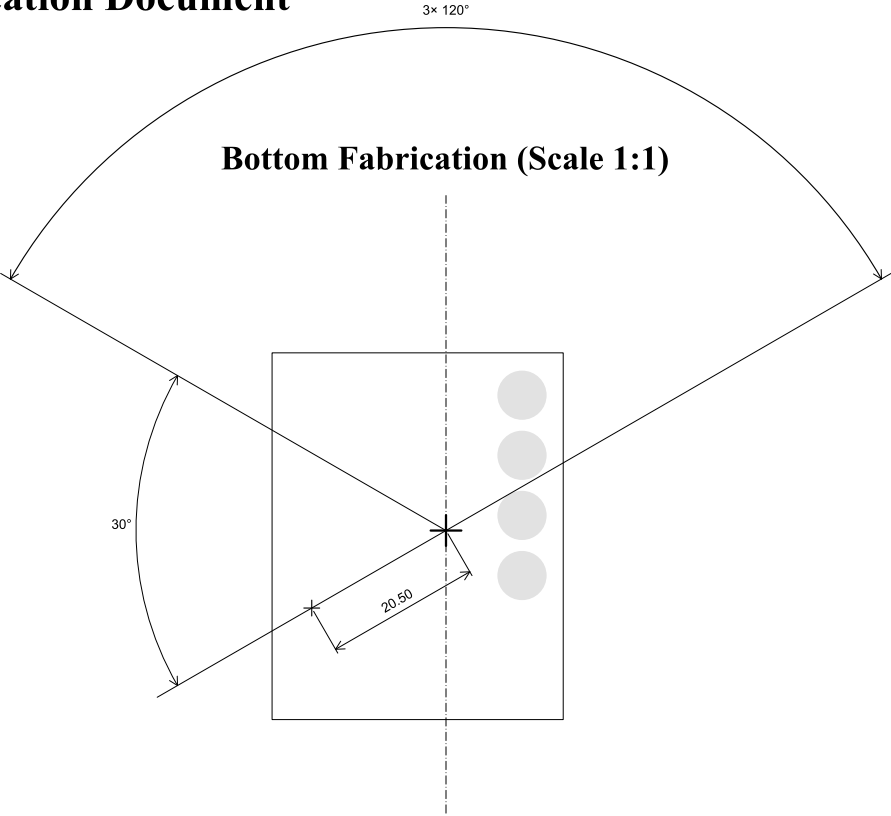
COPPER TO EDGE0,250 mm

HOLE TO HOLE0,254 mm
- 11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: Datalogic Vietnam		Variant: PRELIMINARY	Git Hash: c3c7f22
		Board Name: HAT_RASPI5_INA226		Project Name: HAT INA226	
	Sheet Title: Top Fabrication (Scale 1:1)	File Name: 01_HAT_INA226.kicad_pcb	Designer: ThuongNguyen	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 1 of 11

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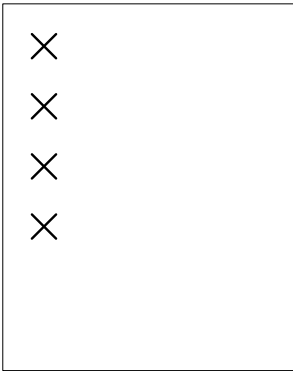
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		Board Name: HAT_RASPI5_INA226		Project Name: HAT INA226	
	Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: 01_HAT_INA226.kicad_pcb	Designer: ThuongNguyen	Date: 2024-04-13	Revision: + (Unreleased)
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	3.20mm (125,98mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
	Total 4					

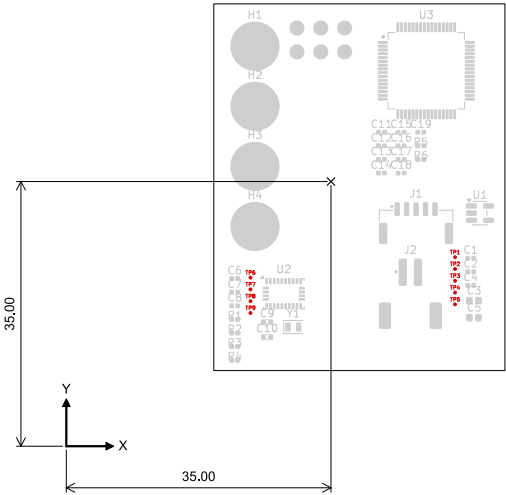
Drill Drawing L1 - L6 (Scale 1:1)



	Comments:	Company: Datalogic Vietnam		Variant: PRELIMINARY	Git Hash: e3c7f22
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	Sheet Title: Drill Drawing (L1 - L6)	File Name: 01_HAT_INA226.kicad_pcb	Designer: ThuongNguyen	Date: 2024-04-13	Revision: + (Unreleased)
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Top Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
TP1	SWDIO	100.90	6.00
TP2	SWCLK	100.90	4.44
TP3	NRST	100.90	2.88
TP4	+3V3	100.90	1.32
TP5	GND	100.90	-0.24
TP6	IMU_INT	73.85	3.30
TP7	IMU_RESET	73.85	1.74
TP8	I2C2_SDA	73.85	0.18
TP9	I2C2_SCL	73.85	-1.38

Ref.	Net	X [mm]	Y [mm]
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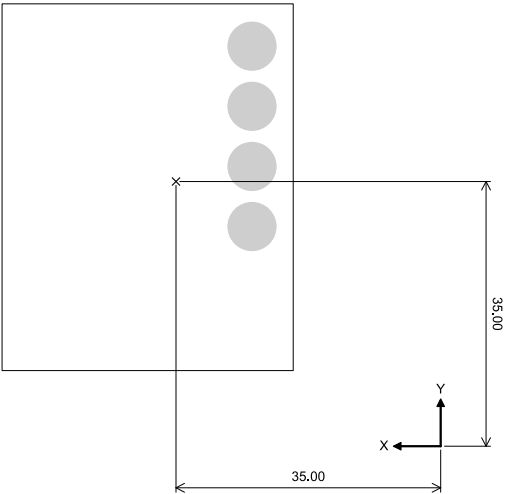
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	Sheet Title: Top Test Points (Scale 1:1)	File Name: 01_HAT_INA226.kicad_pcb	Designer: ThuongNguyen	Date: 2024-04-13	Revision: + (Unreleased)
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Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
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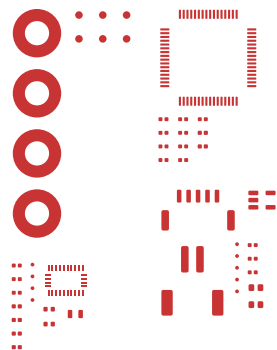


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L1 (Sig, PWR) (Scale 1:1)



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	Sheet Title: L1 (Sig, PWR) (Scale 1:1)	File Name: 01_HAT_INA226.kicad_pcb	Designer: ThuongNguyen	Date: 2024-04-13	Revision: + (Unreleased)
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L2 (GND) (Scale 1:1)



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	Sheet Title: L2 (GND) (Scale 1:1)	File Name: 01_HAT_INA226.kicad_pcb	Designer: ThuongNguyen	Date: 2024-04-13	Revision: + (Unreleased)
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L3 (Sig, PWR) (Scale 1:1)



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		Board Name: HAT_RASPI5_INA226		Project Name: HAT INA226	
	Sheet Title: L3 (Sig, PWR) (Scale 1:1)	File Name: 01_HAT_INA226.kicad_pcb	Designer: ThuongNguyen	Date: 2024-04-13	Revision: + (Unreleased)
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L4 (Sig, PWR) (Scale 1:1)



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		HAT_RASPI5_INA226		HAT INA226	
	L4 (Sig, PWR) (Scale 1:1)	File Name:	Designer:	Date:	Revision:
		01_HAT_INA226.kicad_pcb	ThuongNguyen	2024-04-13	+ (Unreleased)
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L5 (GND) (Scale 1:1)



	Comments:	Company: Datalogic Vietnam		Variant: PRELIMINARY	Git Hash: e3c7f22
		Board Name: HAT_RASPI5_INA226		Project Name: HAT INA226	
	Sheet Title: L5 (GND) (Scale 1:1)	File Name: 01_HAT_INA226.kicad_pcb	Designer: ThuongNguyen	Date: 2024-04-13	Revision: + (Unreleased)
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L6 (Sig, PWR) (Scale 1:1)



	Comments:	Company:		Variant:	Git Hash:
		Datalogic Vietnam		PRELIMINARY	c3c7f22
	Sheet Title:	Board Name:		Project Name:	
		HAT_RASPI5_INA226		HAT INA226	
	L6 (Sig, PWR) (Scale 1:1)	File Name:	Designer:	Date:	Revision:
		01_HAT_INA226.kicad_pcb	ThuongNguyen	2024-04-13	+ (Unreleased)
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